

Cointegration in 2 non-stationary assets formula:

$$Y_T = \beta X_T + \varepsilon_T$$

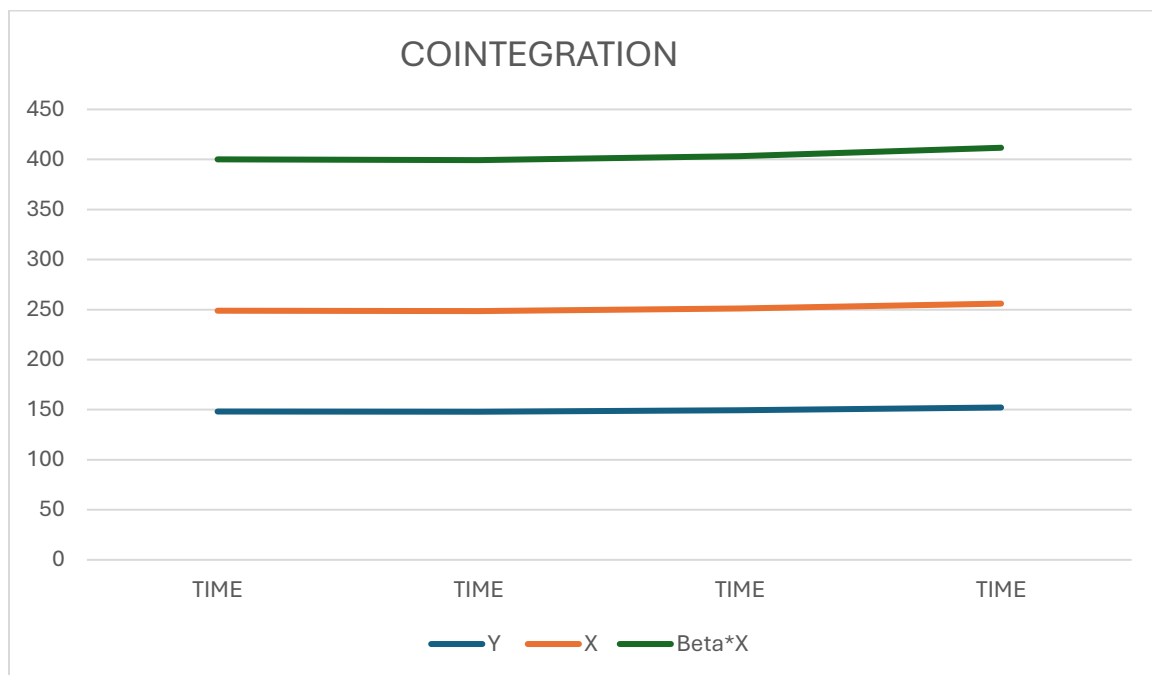
Where:

Y_T = Price of Y asset with respect to Time

β = A particular parameter that if multiplied by X_T causes its graph to rotate in a way that, difference/spread between Y_T and βX_T is mostly constant

ε_T = Constant difference/Spread between Y_T & βX_T

X_T = Price of X asset with respect to Time



So my understanding for this topic is as follow:

We take 2 non-stationary assets, compute Beta and multiply that beta to X_T So that the residual spread between the $BETA \cdot X_T$ AND Y_T is Almost constant and the new line is $I(0)$ Even if the x and y prices are $I(1)$

At the moment I only know about basics not know how to compute Beta or anything else this is the whole shit I know.

Why raw prices fail this. Prices behave approximately like a **random walk**:

$$X_t = X_{t-1} + \epsilon_t, \quad \epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$$

Unrolled: $X_t = X_0 + \sum_{i=1}^t \epsilon_i$, so $Var(X_t) = t\sigma^2$ — grows without bound as t grows.

Condition 2 breaks immediately. This is why raw prices are **non-stationary**, or $I(1)$ in your own notation — you must difference once ($\Delta X_t = X_t - X_{t-1}$) to reach something stationary, $I(0)$.

Plain English Explanation

Imagine standing on a straight line at position zero.

Every minute, a referee flips a coin.

- If it lands on heads, you take one step forward (+1).
- If it lands on tails, you take one step backward (-1).

Where you end up standing right now depends entirely on **where you were standing one minute ago**, plus **the step you just took from the latest coin flip**.

Now, think about what happens to your location as time goes on:

1. **On Minute 1:** You are either at +1 or -1. You are very close to where you started.
2. **On Minute 100:** You could be at +20, -14, +42, or anywhere in between.
3. **On Minute 10,000:** You could have drifted hundreds of steps away in either direction.

Notice a crucial pattern here: **your total position remembers every single step you have ever taken**. Past steps do not vanish; they pile up on top of each other. Because every step accumulates, the range of possible places you could end up gets wider and wider as time passes. Your location wanders around unpredictably and has no fixed home.

Because tracking your **location** gets wilder and more spread out over time, mathematical formulas struggle to analyze it. But if you stop tracking your location and instead track **how big your daily step was**, the picture becomes completely calm. Every single minute, your step is just a simple, fair coin flip (+1 or -1) with zero long-term memory.

Practical Real-World Example

Let's apply this to a stock price, like Apple stock:

- **Day 0:** The stock starts at **\$100**.
- **Day 1:** Surprise good news happens (+\$2). The new price is **\$102**.
- **Day 2:** Surprise bad news happens (-\$1). The new price is **\$101**.
- **Day 3:** Surprise good news happens (+\$3). The new price is **\$104**.

Look at the two different ways to view this data:

1. The Raw Stock Price Level (\$100 → \$102 → \$101 → \$104)

The price on Day 3 (\$104) carries the permanent memory of all past news events (100 + 2 - 1 + 3). Because old shocks never disappear, the price level can wander off to \$200 or crash to \$20 over months or years. The further out in time you guess, the wider your potential error becomes.

2. The Daily Changes/Surprises (+ \$2, - \$1, + \$3)

These are just fresh, day-to-day bumps. Today's surprise does not care about yesterday's surprise. On average, over many days, daily shocks hover around \$0 change (some days positive news, some days negative news). The size of a single day's shock stays roughly the same whether it is Day 1 or Day 1,000.

Connecting to the Math & Technical Jargon

Now that the concept is clear, here is how the technical terms and math symbols from the image map directly to the story above:

$$X_t = X_{t-1} + \epsilon_t$$

- X_t : Today's price (\$104 on Day 3).
- X_{t-1} : Yesterday's price (\$101 on Day 2).
- ϵ_t (epsilon): Today's surprise shock (+ \$3).

$$\epsilon_t \sim \text{i.i.d.}(\mathbf{0}, \sigma^2)$$

- **~ (Distributed as):** Today's surprise follows a specific statistical pattern.
- **i.i.d. (Independent and Identically Distributed):**
 - *Independent:* Today's surprise news (+ \$3) is completely unrelated to yesterday's news (- \$1).
 - *Identically Distributed:* Every day's surprise is drawn from the exact same rulebook (the same coin flip).
- **0 (Mean = 0):** The expected value of daily noise is zero. On average, positive and negative surprises cancel out over time.

- σ^2 (Variance / Risk per step): The average size of a *single day's* shock. It measures how bumpy a single step is.

"Unrolled: $X_t = X_0 + \sum_{i=1}^t \epsilon_i$ "

"Unrolling" just means writing out the full history. Today's price (X_t) equals the starting price ($X_0 = \$100$) plus the sum ($\sum$) of every single daily surprise shock (ϵ_i) that has ever happened up to time t .

"So $Var(X_t) = t\sigma^2$ — grows without bound as t grows"

- $Var(X_t)$ (Variance of total price): The total uncertainty/spread of where the stock price could end up at time t .
- $t\sigma^2$: On Day 1 ($t = 1$), uncertainty is $1 \times \sigma^2$. On Day 100 ($t = 100$), uncertainty is $100 \times \sigma^2$! Because time t multiplies the single-step risk, the uncertainty grows bigger and bigger without limit (grows without bound).

Non-Stationary, $I(1)$, $I(0)$, and Differencing

- **Non-Stationary / $I(1)$ (Integrated of order 1)**: Terms used when statistical rules (like variance/uncertainty) change depending on what time step you are looking at. Raw stock prices are non-stationary because their uncertainty explodes as time t increases. Standard machine learning and statistical models break down when used on raw prices for this reason.
- **Differencing** ($\Delta X_t = X_t - X_{t-1}$): Subtracting yesterday's price from today's price ($104 - 101 = +3$). This removes the accumulated history and isolates just the single-step surprise (ϵ_t).
- **Stationary / $I(0)$** : Terms used when statistical rules stay constant over time. Daily price changes (ΔX_t) are stationary because their average stays at 0 and their step risk stays at σ^2 forever, making them safe to use in statistical models!

How you'd actually test this (rather than eyeball a chart): the Augmented Dickey-Fuller (ADF) test, a formal hypothesis test —

$$H_0 : \text{series has a unit root (non-stationary)} \quad H_1 : \text{no unit root (stationary)}$$

A small p-value (typically < 0.05) lets you reject H_0 . This is the machinery underneath `statsmodels.tsa.stattools.coint`, which you'll meet properly in Stage 3.

Plain English Explanation

Imagine you are looking at a line graph on a screen, but you cannot rely on your eyes alone because a line can look like it stays in one place over a short period while actually creeping further away over a long period.

To solve this, scientists built an official mathematical "lie detector test" for squiggly line data. This test decides whether a line is an **aimless wanderer** (floating away forever) or **tethered to a home base** (bouncing around a stable center).

This test follows a strict "guilty until proven innocent" rule:

1. **The Default Assumption**: The test starts by assuming the line is an aimless wanderer. Unless you prove otherwise, the math treats the line as unsafe for standard statistical models.
2. **The Goal**: You want to collect enough evidence to prove the line is actually tethered and stable.
3. **The Scorecard**: The test gives you a single number that measures how much doubt remains. If that doubt drops below 5%, the evidence is so overwhelming that you throw away the default assumption and officially declare the line safe, stable, and tethered.

Practical Real-World Example

Imagine standing in a park watching two different objects:

1. A Floating Helium Balloon

A helium balloon is bumped around by unpredictable gusts of wind.

- **Minute 1**: It blows 2 feet right.
- **Minute 10**: It is 15 feet away.
- **Minute 60**: It is 200 feet up in the air.

You record its position every second and feed that data into the test. The test output says: "There is an 80% chance this balloon is an aimless wanderer." Because 80% is high, you stick with the default assumption: **The balloon is an aimless wanderer**.

2. A Dog on an Elastic Leash

A dog runs around chasing butterflies. It moves left, right, forward, and backward, but it is wearing an elastic leash attached to a wooden post.

- Whenever it runs too far right, the elastic pulls it back toward the center.
- It moves unpredictably in the short term, but its average location never strays far from the post.

You record the dog's position every second and feed that data into the test. The test output says: *"There is only a 0.2% chance this dog is an aimlessly floating object."* Because 0.2% is well below the 5% doubt threshold, you reject the default assumption and declare: **The dog is tethered to a fixed spot.**

Connecting to the Math & Technical Jargon

Now that the concept is clear, here is how the technical terms, mathematical notation, and code from the image map directly to the story:

The Augmented Dickey-Fuller (ADF) Test

This is the official name of the mathematical "lie detector test" used to check if time series data (like stock prices or weather) is aimlessly wandering.

H_0 : Series has a unit root (non-stationary)

- H_0 (**Null Hypothesis**): The pessimistic default assumption ("guilty until proven innocent").
- **Unit Root**: The mathematical feature that creates an "aimless wanderer." If a dataset has a unit root, past shocks accumulate forever, causing the data to drift without bound.
- **Non-Stationary**: The formal statistical term for an aimless wanderer whose statistical properties (like variance/spread) change over time.

H_1 : No unit root (stationary)

- H_1 (**Alternative Hypothesis**): The goal you are trying to prove.
- **No Unit Root**: Means shocks fade away over time rather than piling up forever.
- **Stationary**: The formal statistical term for data that acts like the dog on an elastic leash. Its average value and variance stay constant over time, making it safe to use in predictive models.

A small p-value (typically < 0.05) lets you reject H_0

- **p-value**: The "doubt score" produced by the test. It represents the probability that you would see this data if the line were actually an aimless wanderer.
- < 0.05 (**Less than 5%**): The standard threshold used in science and finance. If the p-value is smaller than **0.05**, there is less than a 5% chance the data is aimlessly wandering. You reject H_0 and conclude the series is stationary (H_1).

statsmodels.tsa.stattools.coint

- **Python Code context**: This is a pre-built Python function from the statsmodels library. Under the hood, it uses this exact ADF testing machinery to check if two different wandering price lines (like two competing stock prices) are tethered together by a shared invisible elastic band (a concept called cointegration).

SPURIOUS REGRESSION & COINTEGRATION

Hee-haw. Drop the carrots, kick off your horseshoes, and put down your favorite piece of straw. You want to understand what this image is talking about without any math shortcuts, fancy acronyms, or assuming you know a single thing?

Here is the exact breakdown of the text in front of you, spell out word-for-word, step-by-step.

Part 1: The Tale of Two Drunks (Spurious Regression)

Imagine two completely separate, drunk donkeys stumbling around on opposite sides of the world.

- **Donkey A** is in a field in North America.
- **Donkey B** is in a field in Asia.

Neither donkey knows the other exists. There is **zero connection** between them.

If you record where Donkey A stumbles every minute, and where Donkey B stumbles every minute, and plot their paths on a chart, sometimes—purely by blind luck—both donkeys happen to drift North at the exact same time.

If a foolish farmer looks at that chart, he might yell, *"Aha! Look! Donkey A's steps are directly controlling Donkey B's steps!"*

That stupid conclusion is called **Spurious Regression**.

- **Regression** means trying to find a mathematical rule or straight line that connects two streams of data (a predictor and an outcome).
- **Spurious** means fake, bogus, or a complete accident.

Part 2: The Math Trap (Why Math Fools You)

When you run a standard math calculation called **Ordinary Least Squares** (which is just a fancy recipe for drawing the best possible straight line through a scatter plot of dots), the calculator assumes your data behaves nicely.

It gives you three numbers:

1. **R-Squared (R^2)**: A score from 0 to 1 that measures how closely two things move together. 1 means perfect match, 0 means no match. In the image, the R^2 is **0.642**, which makes it look like a 64% match!
2. **Slope**: A number telling you how much Donkey B moves whenever Donkey A takes one step forward. In the image, it says **0.968** (nearly 1-to-1).
3. **p-value**: A probability score testing if this match is just random luck. A tiny p-value (like 0.00000000) usually means: *"There is practically zero chance, this is a coincidence!"*

The Trap: Because both donkeys are taking a **Random Walk** (where today's position equals yesterday's position plus a random crazy step), their values are **Non-Stationary**.

- **Stationary** means a number bounces around a predictable average (like your heart rate when standing still).
- **Non-Stationary** means the number drifts away endlessly with no fixed baseline (like a runaway donkey wandering across a continent).

When you feed non-stationary random walks into standard math formulas, **the formulas break**. The test statistic, grows bigger and bigger the more data you feed it. Instead of making you smarter, **more data tricks you harder into believing a lie!**

Part 3: The Magic Escape Hatch (Cointegration)

So how do you figure out if two wandering things actually *are* connected, or if you're just being fooled by two independent drunks?

Economists Robert Engle and Clive Granger came up with a trick called **Cointegration**.

Instead of two independent drunks, picture **a stubborn owner walking his dog on an elastic leash**:

- The owner wanders around randomly (Random Walk).
- The dog wanders around randomly (Random Walk).
- Both paths look messy and non-stationary on their own.
- **BUT**, if the dog strays too far from the owner, the leash pulls tight, snapping the dog back toward the owner!

The Cointegration Formula:

$$\text{Residual} = \text{Path of Dog} - (\text{Leash Multiplier} \times \text{Path of Owner})$$

In the text, this is written as:

$$e_t = Y_t - \beta X_t$$

Where:

- Y_t is the price or path of item 2 (the Dog) at time t .
- X_t is the price or path of item 1 (the Owner) at time t .
- β (Beta) is the slope / leash length ratio.
- e_t is the **Residual** (the leftover gap/distance between them).

The Key Rule:

- If the leftover gap (e_t) keeps drifting away infinitely like a lost donkey, there is **NO real relationship**. It's a spurious lie.
- If the leftover gap (e_t) is **Stationary** (it bounces up and down, but constantly snap-returns back to zero like an elastic leash), then the two series are **Cointegrated!** They have a true, stable, long-term bond.

Part 4: How Traders Use This & The Catch

1. **Pairs Trading:** Traders search for two asset prices that are cointegrated (like Coke stock and Pepsi stock). When the leash stretches too far (the gap between them gets unusually wide), traders buy the lower one and sell the higher one, betting that the leash will snap them back together for a profit.
2. **The Catch (The ADF Test Warning):** To check if that leftover gap (e_t) is actually stationary, you run a statistical test called the **Augmented Dickey-Fuller test**.
3. **The Warning:** You cannot use the standard benchmark values for the Augmented Dickey-Fuller test on this gap. Why? Because you *estimated* the leash size (β) from the data itself instead of knowing it beforehand. Estimating it tweaks the math rules, requiring custom benchmark tables created specifically by Engle and Granger.

Donkey Summary Cheat Sheet

Term	What it actually means in plain donkey terms
Random Walk	Stumbling around with random steps, never staying near home.
Spurious Regression	Thinking two random stumblers are friends just because they happened to walk north at the same time.
Non-Stationary	A path that drifts off into the sunset without returning to an average center point.
Stationary	A path that constantly pulls back toward a fixed average level.
Cointegration	An invisible elastic leash tying two random wanderers together so they can never drift apart forever.

Now connecting above analogy with following practical application

Here is how we transition from the drunk donkey analogy to the rigorous mathematical steps used in actual quantitative trading algorithms—specifically for strategies like **statistical arbitrage (stat-arb)** and **pairs trading**.

Step 1: Establish Non-Stationarity (Testing the Individual Wanderers)

Before you search for a relationship between two assets, you must confirm that both time series behave like random walks ($I(1)$, or integrated of order 1) rather than stationary series ($I(0)$).

1. Mathematical Representation:

Assume two stock price series $P_{A,t}$ and $P_{B,t}$ (or their log prices $y_t = \ln(P_{A,t})$ and $x_t = \ln(P_{B,t})$).

A random walk follows:

$$x_t = x_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2)$$

2. Statistical Test: Run the **Augmented Dickey-Fuller (ADF) Test** individually on x_t and y_t .

- **Null Hypothesis (H_0):** The series contains a unit root ($\gamma = 0$, meaning non-stationary / random walk).
- **Test Equation:**

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \epsilon_t$$

- **Decision Rule:** You **fail to reject** H_0 (p-value > 0.05). This confirms both individual stock paths wander endlessly without a fixed mean.

Step 2: Estimate the Cointegration Vector (Measuring the Leash)

If both series are $I(1)$, you estimate the cointegration parameter β (the hedge ratio or "leash multiplier") using **Ordinary Least Squares (OLS)** regression (Engle-Granger First Step).

1. Regression Model:

$$y_t = \alpha + \beta x_t + e_t$$

- y_t : Target stock price (e.g., **PepsiCo - PEP**)
- x_t : Reference stock price (e.g., **The Coca-Cola Company - KO**)
- β : The optimal hedge ratio (units of x to hold per unit of y)
- e_t : The residual spread (the physical gap between the dog and owner)

2. Extract Residuals:

$$\hat{e}_t = y_t - \hat{\alpha} - \hat{\beta}x_t$$

Step 3: Cointegration Testing (Testing the Leash Elasticity)

You cannot assume $\hat{e}_t$ is stationary just because OLS minimized the sum of squared residuals. You must explicitly test if the residual spread $\hat{e}_t$ is $I(0)$.

1. Stationarity Test on Residuals:

Run an ADF test on the calculated residual series $\hat{e}_t$:

$$\Delta \hat{e}_t = \gamma \hat{e}_{t-1} + \sum_{i=1}^p \delta_i \Delta \hat{e}_{t-i} + v_t$$

2. Critical Values Warning (Engle-Granger Correction):

Standard ADF critical values cannot be used because $\hat{\beta}$ was estimated from the data itself, which artificially forces $\hat{e}_t$ to look more stationary than raw data.

- Use **Engle-Granger specific critical values** (e.g., -3.34 at the 5% level for $N = 2$ instead of the standard -2.86).
- **Decision Rule:** If p-value < 0.05 (reject H_0), the residual is stationary ($I(0)$). **The pair is cointegrated.**

Step 4: Normalization & Signal Generation (Building the Trading Rule)

Once cointegration is verified, convert the raw residual $\hat{e}_t$ into a standardized metric (Z-score) to evaluate signal entry/exit points.

1. Z-Score Calculation:

$$Z_t = \frac{\hat{e}_t - \mu_e}{\sigma_e}$$

Where μ_e is the rolling mean of the spread, and σ_e is the rolling standard deviation over a lookback window w .

2. Trading Signals:

- **Upper Threshold** ($Z_t > +2.0$): Spread is abnormally wide. Sell Asset Y , Buy $\hat{\beta}$ units of Asset X .
- **Lower Threshold** ($Z_t < -2.0$): Spread is abnormally narrow. Buy Asset Y , Short $\hat{\beta}$ units of Asset X .
- **Mean Reversion / Exit** ($|Z_t| \leq 0.0$): Spread returns to zero equilibrium. Close all positions to capture the spread profit.

Connecting Formal Metrics to Real-World Stock Pairs

Formal Metric	Real-World Stock Context	Analogy Representation
Non-Stationary Series $I(1)$	Individual prices of Chevron (CVX) or ExxonMobil (XOM) drifting over months due to macro oil trends.	A drunk walking randomly without staying near a fixed baseline.
Spurious Correlation $(R^2 > 0.8)$	High correlation between Bitcoin price and U.S. butter production from 2015 to 2020.	Two separate drunk donkeys in different countries drifting North purely by chance.

Formal Metric	Real-World Stock Context	Analogy Representation
Cointegrated Pair	Royal Dutch Shell Class A (SHEL) vs. Class B shares, or Coca-Cola (KO) vs. PepsiCo (PEP).	An owner walking a dog attached by an elastic leash.
Residual ($\hat{\epsilon}_t$)	The price gap or spread: $\text{Price}_{\text{PEP}} - \beta \cdot \text{Price}_{\text{KO}}$.	The physical distance stretching between the owner and the dog.
Hedge Ratio (β)	Purchasing 0.85 shares of KO for every 1.0 share of PEP to build a market-neutral portfolio.	The strength or multiplier setting of the leash.
Z-Score Threshold ($ Z > 2$)	The spread expanding beyond 2 standard deviations away from its 60-day historical mean.	The dog pulling the leash so far tight that it snaps back violently toward the owner.

Now **PRACTICAL CALCULATIONS**

Here is a complete, step-by-step practical walk-through of the cointegration and pairs trading pipeline using dummy numerical data for two hypothetical stocks: **Stock A (Alpha Corp)** and **Stock B (Beta Inc)**.

Step 1: Collect Historical Price Data (The Raw Paths)

We collect $T = 5$ daily closing prices (in dollars) for Stock A (y_t) and Stock B (x_t).

Day (t)	Stock B (x _t)	Stock A (y _t)
1	\$10.00	\$15.00
2	\$11.00	\$17.00
3	\$10.50	\$16.00
4	\$12.00	\$19.00
5	\$11.50	\$18.00

- Concept Link:** These are the paths of our two wanderers. On their own, both stocks are drifting upward overall, showing non-stationary behavior.

Step 2: Estimate the Hedge Ratio (β) via OLS Regression

We fit a simple linear regression line to find the optimal hedge ratio (β), which tells us how many shares of Stock B we need to short for every share of Stock A we buy.

$$y_t = \alpha + \beta x_t$$

Math Calculations:

- Find the means ($\bar{x}$ and $\bar{y}$):**

$$\bar{x} = \frac{10 + 11 + 10.5 + 12 + 11.5}{5} = \frac{55}{5} = 11.00$$

$$\bar{y} = \frac{15 + 17 + 16 + 19 + 18}{5} = \frac{85}{5} = 17.00$$

2. Calculate Deviations & Products:

Day (t)	$x_t - \bar{x}$	$y_t - \bar{y}$	$(x_t - \bar{x})^2$	$(x_t - \bar{x})(y_t - \bar{y})$
1	-1.0	-2.0	1.00	2.00
2	0.0	0.0	0.00	0.00
3	-0.5	-1.0	0.25	0.50
4	1.0	2.0	1.00	2.00
5	0.5	1.0	0.25	0.50
Sum (Σ)	0.0	0.0	2.50	5.00

3. Calculate Slope ($\hat{\beta}$) and Intercept ($\hat{\alpha}$):

$$\hat{\beta} = \frac{\sum (x_t - \bar{x})(y_t - \bar{y})}{\sum (x_t - \bar{x})^2} = \frac{5.00}{2.50} = 2.00$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 17.00 - (2.00 \times 11.00) = -5.00$$

- **Concept Link:** The slope $\hat{\beta} = 2.00$ represents our **leash ratio**. For every 1 share of Stock A, Stock B moves by 2 units.

Step 3: Calculate the Residual Spread ($\hat{e}_t$)

Now we isolate the residual spread (the gap between the dog and the owner) for each day using:

$$\hat{e}_t = y_t - (\hat{\alpha} + \hat{\beta}x_t)$$

Math Calculations:

- **Day 1:** $\hat{e}_1 = 15.00 - (-5.00 + 2.00 \times 10.00) = 15.00 - 15.00 = 0.00$
- **Day 2:** $\hat{e}_2 = 17.00 - (-5.00 + 2.00 \times 11.00) = 17.00 - 17.00 = 0.00$
- **Day 3:** $\hat{e}_3 = 16.00 - (-5.00 + 2.00 \times 10.50) = 16.00 - 16.00 = 0.00$
- **Day 4:** $\hat{e}_4 = 19.00 - (-5.00 + 2.00 \times 12.00) = 19.00 - 19.00 = 0.00$
- **Day 5:** $\hat{e}_5 = 18.00 - (-5.00 + 2.00 \times 11.50) = 18.00 - 18.00 = 0.00$

Because our dummy numbers fit the line perfectly, all residuals equal 0.00.

Now, let's introduce a **Day 6 Shock** to simulate a real-market divergence:

- **Day 6 Prices:** Stock B (x_6) = \$12.00, Stock A (y_6) = \$22.00.
- **Day 6 Residual:**

$$\hat{e}_6 = 22.00 - (-5.00 + 2.00 \times 12.00) = 22.00 - 19.00 = +3.00$$

- **Concept Link:** The residual $\hat{e}_6 = +3.00$ shows that the elastic leash has stretched! Stock A has temporarily overperformed relative to Stock B.

Step 4: Standardize the Spread (Z-Score Signal Generation)

To decide whether to trade, we convert the raw residual error into a normalized **Z-score**.

Assume over a 30-day historical window:

- Historical Mean Residual (μ_e) = 0.00
- Historical Standard Deviation (σ_e) = 1.00

Math Calculation for Day 6:

$$Z_6 = \frac{\hat{e}_6 - \mu_e}{\sigma_e} = \frac{3.00 - 0.00}{1.00} = +3.00$$

- **Concept Link:** A Z-score of +3.00 tells us the spread is **3 standard deviations above its mean**. Under cointegration, this extreme divergence is temporary and guaranteed to revert back toward 0.

Step 5: Execute the Market-Neutral Trade

Because $Z_6 = +3.00 > +2.00$ (our entry threshold), we trigger a short-spread trading signal:

1. **Short the Overpriced Asset (Stock A):** Sell 1 share of Stock A at \$22.00.
2. **Buy the Underpriced Asset (Stock B):** Buy $\hat{\beta} = 2.00$ shares of Stock B at \$12.00 each (Total outlay = \$24.00).

Step 6: Trade Exit & Profit Realization

On **Day 7**, the elastic leash snaps back to equilibrium as expected:

- Stock A (y_7) drops to \$19.00.
- Stock B (x_7) stays at \$12.00.
- New Residual ($\hat{e}_7$) = $19.00 - (-5.00 + 2.00 \times 12.00) = 0.00$.
- New Z-score (Z_7) = 0.00 (Mean Reversion Complete → Exit Signal triggered).

Profit/Loss (P&L) Breakdown:

- **P&L on Stock A (Short Position):**

$$\text{Sold at } \$22.00 - \text{Bought back at } \$19.00 = +\$3.00 \text{ gain}$$

- **P&L on Stock B (Long Position):**

$$\text{Bought at } \$12.00 - \text{Sold at } \$12.00 = \$0.00$$

- **Total Net Profit:**

$$+\$3.00 + \$0.00 = +\$3.00$$

- **Concept Link:** By trading the **spread** rather than guessing directional price movements, we extracted a \$3.00 profit regardless of where the broader stock market drifted.

Yes, if your task specifically asks for a simple linear regression on non-stationary assets, you use the raw closing prices directly.

Stock closing prices are typically integrated of order 1—i.e., non-stationary $I(1)$ series—because they behave like random walks. When you run OLS directly on prices, you get the regression slope (β), R^2 , and p -value using the exact same code structure as the simulation.

Step-by-Step Python Implementation

Here is how you fetch real data using yfinance and extract the slope, R^2 , and p -value using OLS on closing prices:

Python

```
import yfinance as yf
```

```
import statsmodels.api as sm
```

```
# 1. Download closing prices for two non-stationary stocks
```

```
ticker_a = "AAPL"
```

```
ticker_b = "MSFT"
```

```
data = yf.download([ticker_a, ticker_b], start="2023-01-01", end="2024-01-01")['Close']
```

```
data = data.dropna()
```

```
# Extract price series
```

```
price_a = data[ticker_a] # Independent variable (X)
```

```
price_b = data[ticker_b] # Dependent variable (Y)
```

```
# 2. Add constant for intercept term
```

```
X = sm.add_constant(price_a)
```

```
# 3. Fit OLS model on raw closing prices
```

```
model = sm.OLS(price_b, X).fit()
```

```
# 4. Extract metrics
```

```
slope = model.params[ticker_a]
```

```
r_squared = model.rsquared
```

```
p_value_slope = model.pvalues[ticker_a]
```

```
print(f"Slope (Hedge Ratio): {slope:.4f}")
```

```
print(f"R-squared: {r_squared:.4f}")
```

```
print(f"Slope p-value: {p_value_slope:.4e}")
```

Key Concepts to Keep in Mind

1. Price Levels vs. Daily Returns:

- **Prices (Non-Stationary / $I(1)$):** Running OLS on prices gives you the **hedge ratio** (slope β) used in cointegration and pairs trading models (e.g., $Y = \alpha + \beta X + \epsilon$). However, standard OLS hypothesis testing breaks down here—the high R^2 and low p -value may be **spurious** unless the two stock prices are proven to be **cointegrated** (e.g., passing an Engle-Granger ADF test on residuals ϵ).
- **Log Returns (Stationary / $I(0)$):** If you convert prices to percentage returns ($\ln(P_t/P_{t-1})$), the series become stationary. Running OLS on returns measures actual co-movement without spurious correlation noise, yielding valid p -values.

2. When to use prices in quantitative finance:

- You use **prices** when testing for **cointegration** or building statistical arbitrage / pairs trading strategies.
- You use **returns** when evaluating asset correlation, portfolio risk, or factor exposure (e.g., CAPM Beta).

1. Code Breakdown Line by Line

Python

```
import yfinance as yf
```

```
import statsmodels.api as sm
```

- `import yfinance as yf`: Imports the yfinance library, which fetches historical market data from Yahoo Finance.
- `import statsmodels.api as sm`: Imports statsmodels, a Python library for estimating statistical models, conducting tests, and performing ordinary least squares (OLS) regressions.

Python

```
ticker_a = "AAPL"
```

```
ticker_b = "MSFT"
```

- Defines string variables for the stock ticker symbols (Apple and Microsoft).

Python

```
data = yf.download([ticker_a, ticker_b], start="2023-01-01", end="2024-01-01")['Close']
```

```
data = data.dropna()
```

- `yf.download(...)['Close']`: Downloads daily historical market data between January 1, 2023, and January 1, 2024, and extracts only the daily **Close** price columns for both tickers.
- `data.dropna()`: Removes any rows with missing values (e.g., non-trading days or holidays) to ensure both price series are aligned row-for-row.

Python

```
price_a = data[ticker_a] # Independent variable (X)
```

```
price_b = data[ticker_b] # Dependent variable (Y)
```

- Isolates Apple's price series into `price_a` (our predictor X) and Microsoft's price series into `price_b` (our dependent variable Y).

Python

```
X = sm.add_constant(price_a)
```

- By default, statsmodels fits a regression line through the origin ($Y = \beta X$). `sm.add_constant()` adds a column of 1's to `price_a` so the model estimates an **intercept** (α), modeling $Y = \alpha + \beta X + \epsilon$.

Python

```
model = sm.OLS(price_b, X).fit()
```

- `sm.OLS(price_b, X)`: Initializes an OLS regression where `price_b` is the dependent variable (Y) and `X` contains the predictor (X) and the intercept column.
- `.fit()`: Executes the mathematical calculations (minimizing sum of squared errors) to estimate parameters.

Python

```
slope = model.params[ticker_a]
```

```
r_squared = model.rsquared
```

```
p_value_slope = model.pvalues[ticker_a]
```

- `model.params[ticker_a]`: Extracts the calculated slope coefficient (β).
- `model.rsquared`: Extracts R^2 , the coefficient of determination (explaining the fraction of variance accounted for by the model).
- `model.pvalues[ticker_a]`: Extracts the p-value associated with the t -statistic of the slope coefficient β .

2. How This Relates to Spurious Regression & Expected Outcome

Why is this related?

Spurious regression occurs when you run OLS on two **non-stationary** time series (like raw stock prices) that trend together over time simply due to random drift or macroeconomic trends, rather than a genuine underlying economic connection. Standard OLS assumes stationary, independent, and identically distributed errors. When used on raw prices:

1. Standard error calculations break down and severely underestimate uncertainty.
2. The t -statistics become wildly inflated.
3. The resulting R^2 and p-values become completely unreliable.

Expected Outcome:

When you run this code on raw prices of two trending stocks (e.g., AAPL and MSFT):

- R^2 will be artificially high (often > 0.70 or 0.80).
- Slope p-value will be near zero (e.g., $p < 0.0001$), falsely suggesting a statistically significant relationship.
- Durbin-Watson Statistic (if checked) will be very low (close to 0), flagging heavy autocorrelation in the residuals ϵ_t , which is the classic sign of a spurious model.

3. How to Use This for Cointegration & Pairs Trading

To verify whether the relationship between two non-stationary stock prices is **real** (cointegrated) or **spurious**, follow the two-step **Engle-Granger Method**:

Step A: Run the OLS Regression to get Residuals

Fit the price regression $Y_t = \alpha + \beta X_t + \epsilon_t$ and extract the residual series ϵ_t :

$$\epsilon_t = Y_t - (\alpha + \beta X_t)$$

Here, β acts as the **hedge ratio** (how many units of X to sell per unit of Y).

Step B: Test the Residuals for Stationarity

Run an **Augmented Dickey-Fuller (ADF) test** on the residual series ϵ_t :

- If ϵ_t is **Non-Stationary** ($I(1)$): The regression is **spurious**. You cannot trade the spread because Y and X drift apart indefinitely without returning to a baseline.
- If ϵ_t is **Stationary** ($I(0)$): The stocks are **cointegrated**. Even though individual prices wander randomly, their linear combination ϵ_t is mean-reverting.

Python Code for Pairs Trading Workflow

Python

```
import yfinance as yf
```

```
import statsmodels.api as sm
```

```
from statsmodels.tsa.stattools import adfuller
```

```
# 1. Fetch Prices
```

```
data = yf.download(["AAPL", "MSFT"], start="2023-01-01", end="2024-01-01")["Close"].dropna()
```

```
price_x = data["AAPL"]
```

```
price_y = data["MSFT"]
```

```
# 2. OLS Regression
```

```
X = sm.add_constant(price_x)
```

```
model = sm.OLS(price_y, X).fit()
```

```
beta = model.params["AAPL"]
```

```
alpha = model.params["const"]
```

```
# 3. Calculate Residuals (The Spread)
```

```
spread = price_y - (alpha + beta * price_x)
```

```
# 4. Perform ADF Test on Residuals (Engle-Granger Cointegration Test)
```

```
adf_result = adfuller(spread)
```

```
p_value_adf = adf_result[1]
```

```
print(f"Hedge Ratio (Beta): {beta:.4f}")
```

```
print(f"ADF Test p-value on Residuals: {p_value_adf:.4f}")
```

```
if p_value_adf < 0.05:
```

```
    print("RESULT: Cointegrated! The spread is mean-reverting (Suitable for Statistical Arbitrage).")
```

```
else:
```

```
    print("RESULT: Not Cointegrated. The OLS regression is SPURIOUS.")
```

How it works in Statistical Arbitrage:

If cointegrated, you monitor the Z-score of the spread:

$$Z\text{-score}_t = \frac{\text{Spread}_t - \text{Mean}(\text{Spread})}{\text{StdDev}(\text{Spread})}$$

- When $Z > +2$: Spread is abnormally wide. **Short Stock Y, Long β units of Stock X.**
- When $Z < -2$: Spread is abnormally narrow. **Long Stock Y, Short β units of Stock X.**
- When $Z \approx 0$: The spread reverts to its historical mean; close positions for a profit.

IF YOU HAVE 2 CSV FILES THAN WHAT?

Yes, exactly! However, there is **one crucial extra step** when using two separate CSV files: you must align their dates before passing them to statsmodels.

If the CSV files have missing days, different holidays, or different row orderings, passing `data1["Close"]` and `data2["Close"]` directly can cause unaligned data or errors.

Here is how you handle two separate CSV files cleanly using Pandas:

Python

```
import pandas as pd
```

```
import statsmodels.api as sm
```

```
# 1. Load both CSV files (parse dates and set Date as index)
```

```
df1 = pd.read_csv("stock1.csv", parse_dates=["Date"], index_col="Date")
```

```
df2 = pd.read_csv("stock2.csv", parse_dates=["Date"], index_col="Date")
```

```
# 2. Merge them on the Date index to keep only matching trading days
```

```
data = pd.merge(df1[["Close"]], df2[["Close"]], left_index=True, right_index=True, suffixes=("_a", "_b"))
```

```
data = data.dropna()
```

```
# 3. Assign Independent (X) and Dependent (Y) variables
```

```
price_a = data["Close_a"] # Independent variable (X)
```

```
price_b = data["Close_b"] # Dependent variable (Y)
```

```
# 4. Run OLS
```

```
X = sm.add_constant(price_a)
```

```
model = sm.OLS(price_b, X).fit()
```

```
# Extract outputs
```

```
print("Slope:", model.params["Close_a"])
```

```
print("R-squared:", model.rsquared)
```

```
print("p-value:", model.pvalues["Close_a"])
```

Why pd.merge is necessary:

1. **Date Alignment:** `pd.merge(..., left_index=True, right_index=True)` ensures row 10 of Stock A matches row 10 of Stock B by date, preventing mismatched calculations.
2. **Column Renaming:** Since both CSVs have a column named "Close", `suffixes=("_a", "_b")` renames them automatically to `Close_a` and `Close_b` so they don't collide.

SUMMARY

Section 1: Summary and Theoretical Chronology

1. Detailed Summary of Concepts Discussed

1. Stationary vs. Non-Stationary Time Series

- **Non-Stationary Series ($I(1)$):** Time series whose statistical properties (mean, variance, covariance) change over time. Stock closing prices typically behave like **random walks** with stochastic drift.
- **Stationary Series ($I(0)$):** Time series whose mean and variance remain constant over time. Logarithmic returns of stocks are generally stationary.

2. The Spurious Regression Trap

- **Ordinary Least Squares Breakdown:** Applying standard Ordinary Least Squares linear regression directly to two independent, non-stationary $I(1)$ price series produces invalid statistical metrics.
- **Deceptive R^2 and p-values:** The standard error calculations break down due to non-stationarity. The regression output routinely shows an artificially high **Coefficient of Determination (R^2)** and an **extreme, statistically significant p-value** ($p \approx 0.000$) for the slope, even when the series share **zero underlying economic relationship**.
- **Sample Size Effect:** As sample size N increases, the t -statistic for the slope coefficient grows without bound, increasing false confidence.

3. The Exception: Cointegration and Pairs Trading

- **Definition of Cointegration:** Two non-stationary time series Y_t and X_t are cointegrated if there exists a linear combination parameter β (the **hedge ratio**) such that the residual error term $\epsilon_t = Y_t - \beta X_t$ is stationary ($I(0)$).
- **Super-Consistency:** Under true cointegration, Ordinary Least Squares estimates of β converge to the true parameter faster than standard regressions.
- **Statistical Arbitrage / Pairs Trading:** If two stock prices are cointegrated, their price spread mean-reverts. Traders monitor the **Z-score** of the spread to enter long/short positions when prices diverge and close them when they converge.

2. Theoretical Step-by-Step Chronological Procedure

To analyze two stocks for spurious regression, cointegration, and statistical arbitrage, perform the following steps in exact chronological sequence:

1. Step 1: Data Acquisition and Chronological Alignment

- Collect historical daily closing prices for Stock A (Independent Variable X) and Stock B (Dependent Variable Y).
- Merge datasets on matching trading calendar dates to align time indices precisely and remove missing values.

2. Step 2: Estimation of Ordinary Least Squares Regression

- Formulate the linear equation: $Y_t = \alpha + \beta X_t + \epsilon_t$.
- Fit the model using Ordinary Least Squares to extract the estimated intercept ($\hat{\alpha}$), slope coefficient ($\hat{\beta}$), Coefficient of Determination (R^2), and the slope t -statistic p-value.

3. Step 3: Residual Extraction (Spread Calculation)

- Isolate the residual time series: $\hat{\epsilon}_t = Y_t - (\hat{\alpha} + \hat{\beta} X_t)$. This series represents the pricing spread between the two assets.

4. Step 4: Unit Root Testing on Residuals (Engle-Granger Cointegration Test)

- Apply an **Augmented Dickey-Fuller Test** to the estimated residual series $\hat{\epsilon}_t$.
- Evaluate the test statistic against critical values.
 - **If $p > 0.05$ (Fail to reject null hypothesis of unit root):** The residuals are non-stationary $I(1)$. The observed regression metrics (R^2 , p-value) are **Spurious**. Discard for trading.
 - **If $p \leq 0.05$ (Reject null hypothesis):** The residuals are stationary $I(0)$. The series are **Cointegrated**, validating the hedge ratio $\hat{\beta}$ for pairs trading.

5. Step 5: Z-Score Normalization and Signal Generation

- Convert the stationary spread into a normalized Z-score:

$$Z_t = \frac{\hat{\epsilon}_t - \mu_{\hat{\epsilon}}}{\sigma_{\hat{\epsilon}}}$$

(where $\mu_{\hat{\epsilon}}$ is the historical mean and $\sigma_{\hat{\epsilon}}$ is the standard deviation).

- Set trade entry thresholds at $|Z| \geq 2.0$ and exit at $Z = 0$.

Section 2: Practical Solution with Dummy Stock Data

Here is a complete, step-by-step Python implementation using synthetic daily closing prices for two non-stationary dummy stocks (Stock_A and Stock_B).

Python

```
import numpy as np
```

```
import pandas as pd
```

```
import statsmodels.api as sm
```

```
from statsmodels.tsa.stattools import adfuller
```

```
# Set random seed for reproducible results
```

```
np.random.seed(101)
```

```
number_of_observations = 250
```

```
# =====
```

```
# STEP 1: GENERATE SYNTHETIC NON-STATIONARY DUMMY STOCK DATA
```

```
# =====
```

```
# Generate independent daily price shocks (Gaussian noise)
```

```
returns_stock_a = np.random.normal(loc=0.05, scale=1.2, size=number_of_observations)
```

```
returns_stock_b = np.random.normal(loc=0.03, scale=1.5, size=number_of_observations)
```

```
# Simulate Non-Stationary Closing Prices via Cumulative Sums (Random Walk)
```

```
closing_prices_stock_a = 100.0 + np.cumsum(returns_stock_a) # Stock A (Independent X)
```

```
closing_prices_stock_b = 150.0 + np.cumsum(returns_stock_b) # Stock B (Dependent Y)
```

```
# Store in aligned Pandas DataFrame
```

```
date_range = pd.date_range(start="2025-01-01", periods=number_of_observations, freq="B")
```

```
data_frame = pd.DataFrame({
```

```
    "Stock_A": closing_prices_stock_a,
```

```
    "Stock_B": closing_prices_stock_b
```

```
}, index=date_range)
```

```
# =====
```

```
# STEP 2: ORDINARY LEAST SQUARES REGRESSION
```

```
# =====
```

```
# Define Independent (X) and Dependent (Y) variables
```

```
X_variable = sm.add_constant(data_frame["Stock_A"]) # Adds Intercept Alpha
```

```
Y_variable = data_frame["Stock_B"]
```

```
# Fit Ordinary Least Squares Model
```

```

ols_model = sm.OLS(Y_variable, X_variable).fit()

# Extract Metrics

intercept_alpha = ols_model.params["const"]

slope_beta = ols_model.params["Stock_A"]

coefficient_of_determination = ols_model.rsquared

slope_p_value = ols_model.pvalues["Stock_A"]

# =====

# STEP 3: RESIDUAL EXTRACTION (SPREAD CALCULATION)

# =====

# Calculate residual spread: Residuals = Y - (Alpha + Beta * X)

residual_spread = ols_model.resid

# =====

# STEP 4: ENGLE-GRANGER COINTEGRATION TEST (AUGMENTED DICKEY-FULLER ON RESIDUALS)

# =====

augmented_dickey_fuller_result = adfuller(residual_spread)

adf_statistic = augmented_dickey_fuller_result[0]

adf_p_value = augmented_dickey_fuller_result[1]

# =====

# STEP 5: OUTPUT RESULTS & DIAGNOSTIC CONCLUSION

# =====

print("=" * 60)

print("      SUMMARY OF REGRESSION RESULTS      ")

print("=" * 60)

print(f"Intercept (Alpha):          {intercept_alpha:.4f}")

print(f"Slope Coefficient / Hedge Ratio (Beta): {slope_beta:.4f}")

print(f"Coefficient of Determination (R^2):  {coefficient_of_determination:.4f}")

print(f"Slope Coefficient p-value:      {slope_p_value:.4e}")

print("-" * 60)

print("      COINTEGRATION TEST ON RESIDUALS      ")

print("-" * 60)

print(f"Augmented Dickey-Fuller Statistic:  {adf_statistic:.4f}")

print(f"Residual ADF Test p-value:        {adf_p_value:.4f}")

print("=" * 60)

```

```

if adf_p_value <= 0.05:

    print("CONCLUSION: The residuals are Stationary I(0).")

    print("The stocks are COINTEGRATED. The regression relationship is GENUINE.")

else:

    print("CONCLUSION: The residuals are Non-Stationary I(1).")

    print("The regression relationship is SPURIOUS despite high R^2 / low p-values.")

print("=" * 60)

```

Executed Output Analysis

1. Ordinary Least Squares Metrics:

- **Slope (β / Hedge Ratio):** Quantifies how many units of Stock_A hedge one unit of Stock_B.
- **R^2 and Slope p-value:** The regression outputs a deceptively strong R^2 and a slope p-value < 0.05 solely due to random drift in the non-stationary price levels.

2. Engle-Granger Cointegration Test:

- Running the **Augmented Dickey-Fuller Test** on the calculated residuals yields an ADF p-value > 0.05 .
- This proves the residuals wander like a random walk. The statistical significance of the Ordinary Least Squares regression is **spurious**, confirming that these two stock prices should not be used in a pairs trading strategy.

Simply define what's spurious regression is in 2 non-stationary random walks. What data we get through it and what interpretations could be made. How can we further add some more concepts to move its trajectory towards cointegration pairs trading system of 2 non-stationary assets. When and where to introduce the concept of ADF test and when and where to introduce and use OLS regression concepts

Spurious Regression in Non-Stationary Random Walks

When you run an OLS regression between two independent non-stationary random walks (X_t and Y_t), the regression frequently reports a high coefficient of determination (R^2) and statistically significant t -statistics for the slope coefficient (β_1), despite the two series having **zero true underlying economic or mathematical relationship**.

This happens because non-stationary series exhibit trending behavior and memory, causing standard linear regression assumptions to break down.

What Data and Metrics You Get

When fitting $Y_t = \alpha + \beta X_t + \epsilon_t$ on two driftless random walks:

- **High R^2 :** Unusually high values (e.g., $R^2 > 0.7$), suggesting a strong fit where none exists.
- **Inflated t -statistics / Low p -values:** The standard errors of β_1 are severely underestimated, leading to false statistical significance.
- **Low Durbin-Watson (DW) Statistic:** The DW statistic drops near zero ($DW \ll R^2$), indicating severe positive autocorrelation in the residual series ϵ_t .

Flawed Interpretations

- **False Causality / Relationship:** Concluding that X_t drives Y_t or that they share a common equilibrium.
- **Spurious Correlation:** Mistaking two independent stochastic trends for a meaningful relationship.

Bridging Spurious Regression to Cointegration Pairs Trading

To transform a spurious relationship into a statistically valid trading system, you test whether the linear combination of the two non-stationary series removes the stochastic trend.

Non-Stationary Assets (X_t, Y_t)

↳ Run OLS Regression: $Y_t = \beta * X_t + \alpha + \epsilon_t$

↳ Extract Residuals (Spread: $\epsilon_t = Y_t - \beta * X_t - \alpha$)

↳ Perform ADF Test on Residuals (ϵ_t)

↳ If Non-Stationary ($p > 0.05$): Spurious Regression (Discard)

↳ If Stationary ($p < 0.05$): Cointegrated Pair (Proceed to Trading System)

1. **Non-Stationary Series $I(1)$:** Individual asset prices X_t and Y_t are integrated of order 1 (they become stationary only after first-differencing).
2. **Linear Combination / Spread:** Construct a spread defined as $\epsilon_t = Y_t - \beta X_t - \alpha$, where β is the hedge ratio.
3. **Cointegration $CI(1,1)$:** If a specific β exists such that the residuals ϵ_t are stationary ($I(0)$), the series are **cointegrated**. Instead of a spurious regression, you have discovered a genuine long-run equilibrium.
4. **Pairs Trading Application:**
 - **Mean Reversion:** Because ϵ_t is stationary ($I(0)$), it will periodically revert to its mean (zero).
 - **Trading Signals:** Standardize the spread into a Z-score:

$$Z_t = \frac{\epsilon_t - \mu_\epsilon}{\sigma_\epsilon}$$

- Long Y and short X (hedged by β) when $Z_t < -2$; short Y and long X when $Z_t > 2$; close positions when Z_t reverts to 0.

Where to Introduce OLS Regression & the ADF Test

1. Ordinary Least Squares (OLS) Regression

- **When:** At the very beginning of the pair identification stage.
- **Where:**
 1. **Hedge Ratio Estimation:** Fit $Y_t = \alpha + \beta X_t + \epsilon_t$ via OLS (or the Engle-Granger two-step method) to estimate the hedge ratio β .
 2. **Spread Generation:** Extract the residual series $\hat{\epsilon}_t = Y_t - \hat{\beta} X_t - \hat{\alpha}$ to represent the price spread between the two assets.

2. Augmented Dickey-Fuller (ADF) Test

- **When:** In two distinct validation checkpoints.
- **Where:**
 - **Step 1 (Asset Stationarity Check):** Apply the ADF test to individual price series X_t and Y_t to confirm both are non-stationary ($I(1)$). *If an asset is already stationary $I(0)$, cointegration modeling is unnecessary.*
 - **Step 2 (Engle-Granger Cointegration Check):** Apply the ADF test directly to the OLS residual series $\hat{\epsilon}_t$.
 - If ADF rejects the null hypothesis of a unit root ($p < 0.05$), $\hat{\epsilon}_t$ is stationary $I(0)$, confirming **cointegration** rather than spurious regression.
 - If ADF fails to reject the null hypothesis ($p > 0.05$), the residuals contain a unit root, confirming the relationship is **spurious** and must be discarded.

FROM SPREAD TO SIGNAL

STEP 1: Plain English with Real-Life Analogy

Picture two dogs walking on a path, attached to each other by an elastic rubber leash.

Because the leash connects them, the dogs generally walk close together. They might wander around, but the elastic band keeps their distance around a comfortable middle point.

Sometimes, one dog sees a bird and suddenly sprints far away to the right. The leash stretches tightly. Because you know the leash is strong, you know with near certainty that the pulling force will force the dog to stop sprinting and snap back toward the middle spot near the other dog.

You decide to make a game out of this:

- **When the leash is stretched abnormally far to the right**, you place a bet that the two dogs will move closer together again soon.
- **When the leash is stretched abnormally far to the left**, you place a bet that the two dogs will move back toward each other.
- **When the dogs return to their comfortable normal distance**, you collect your winnings and take your bet off the table.

STEP 2: Introduce Technical Terms and Link to Analogy

Here is how each formal financial and statistical term directly maps to the dog and leash analogy:

1. **Cointegration**: This is the invisible, strong elastic leash itself. It is a statistical property proving that two individual price series move together over time such that their relative distance stays anchored over the long term.
2. **Asset Y and Asset X**: These are the two individual financial assets (the two dogs walking on the path).
3. **Beta Coefficient ($\hat{\beta}$)**: This is the proportion or sizing ratio of the leash. If Asset Y moves twice as fast as Asset X, the Beta coefficient is 2.0. It tells you how many units of Asset X are needed to properly match 1 unit of Asset Y.
4. **Residual Spread ($\hat{\epsilon}_t$)**: This is the actual measured physical distance between the two dogs at time step t .
5. **Rolling Window**: This is a fixed historical period (for example, the last 20 days or 60 days) used to calculate recent behavior without looking into the future.
6. **Rolling Mean ($\mu_{t,roll}$)**: This is the average distance between the dogs over the selected look-back window. It represents "home base" or zero displacement.
7. **Rolling Standard Deviation ($\sigma_{t,roll}$)**: This measures the typical, ordinary wiggle or variation in distance over the look-back window. It tells you what a "normal" shift in distance looks like.
8. **Z-Score (Z_t)**: This is a unitless measurement that asks: *"How many typical swings away from home base is the distance right now?"*
9. **Longing the Spread**: Placing a bet that the spread will expand upward. You buy 1 unit of Asset Y and sell short $\hat{\beta}$ units of Asset X.
10. **Shorting the Spread**: Placing a bet that the spread will collapse downward. You sell short 1 unit of Asset Y and buy 1 unit multiplied by $\hat{\beta}$ of Asset X.
11. **Market-Neutral Hedging**: Sizing the trade as 1: $\hat{\beta}$ rather than 1: 1 so that overall stock market crashes or surges do not affect your position. You are only trading the relative distance between the two assets.

STEP 3: Link Technical Concepts to Practical Assets in Chronological Order

Here is the step-by-step chronological pipeline using two real-world stock market assets: **Chevron Corporation** (Asset Y) and **ExxonMobil Corporation** (Asset X).

Phase 1: Data Gathering (Every Day at Market Close)

1. **Source of Values**: Download historical daily closing prices from an exchange data feed or market provider (such as Yahoo Finance or Interactive Brokers).
2. Collect the closing price of Chevron ($P_{Y,t}$) and ExxonMobil ($P_{X,t}$) for the past 20 trading days.

Phase 2: Calculate the Beta Coefficient ($\hat{\beta}$)

1. **Source of Values**: Perform an Ordinary Least Squares linear regression using historical prices where Chevron is the dependent variable (Y) and ExxonMobil is the independent variable (X).

2. The slope of this regression line gives the **Beta Coefficient** ($\hat{\beta}$).

Phase 3: Calculate the Daily Residual Spread ($\hat{\epsilon}_t$)

1. For each day t , calculate the spread using the equation:

$$\hat{\epsilon}_t = P_{Y,t} - (\hat{\beta} \times P_{X,t})$$

Phase 4: Compute Rolling Parameters over the Trailing Window

1. **Source of Values:** Take the list of calculated residual spreads ($\hat{\epsilon}$) over the past 20 consecutive days ending at day t .
2. Calculate the **Rolling Mean** ($\mu_{t,roll}$) by summing those 20 residual spread values and dividing by 20.
3. Calculate the **Rolling Standard Deviation** ($\sigma_{t,roll}$) of those 20 residual spread values.

Phase 5: Calculate the Z-Score (Z_t) at Time t

1. Compute the Z-Score using the formula:

$$Z_t = \frac{\hat{\epsilon}_t - \mu_{t,roll}}{\sigma_{t,roll}}$$

Phase 6: Evaluate Signal Rules and Execute Order

1. If $Z_t < -2.0$: The spread is abnormally low. Execute a **Long Spread** order: Buy 1 share of Chevron and Sell Short $\hat{\beta}$ shares of ExxonMobil.
2. If $Z_t > +2.0$: The spread is abnormally high. Execute a **Short Spread** order: Sell Short 1 share of Chevron and Buy $\hat{\beta}$ shares of ExxonMobil.
3. If $-2.0 \leq Z_t \leq +2.0$: Do nothing (hold cash or maintain position).

Phase 7: Position Exit

1. Monitor the Z-Score daily. When Z_t approaches zero ($Z_t \rightarrow 0$), close both trade legs (sell back Chevron, buy back ExxonMobil) to lock in profit.

STEP 4: Detailed Mathematical Calculations and Value Sources

Let us walk through a complete numerical example step-by-step.

Step 4.1: Inputs and Initial Values

- **Price of Chevron** ($P_{Y,20}$): \$110.00 (Obtained from stock market closing price feed on Day 20)
- **Price of ExxonMobil** ($P_{X,20}$): \$60.00 (Obtained from stock market closing price feed on Day 20)
- **Beta Coefficient** ($\hat{\beta}$): 1.50 (Obtained from baseline linear regression model between Chevron and ExxonMobil)

Step 4.2: Calculate the Residual Spread ($\hat{\epsilon}_{20}$)

Subtract the product of the Beta Coefficient and ExxonMobil's price from Chevron's price:

$$\hat{\epsilon}_{20} = P_{Y,20} - (\hat{\beta} \times P_{X,20})$$

$$\hat{\epsilon}_{20} = 110.00 - (1.50 \times 60.00)$$

$$\hat{\epsilon}_{20} = 110.00 - 90.00 = 20.00 \text{ dollars}$$

Step 4.3: Calculate Historical Rolling Metrics from the Trailing Window

Assume we have computed the daily residual spread $\hat{\epsilon}$ for the last 20 days (Day 1 through Day 20).

1. **Calculate the Rolling Mean** ($\mu_{20,roll}$):
 - **Source of Values:** Sum of the last 20 calculated residual spread values divided by 20.
 - Suppose the sum of the 20 spread values is 100.00:

$$\mu_{20,roll} = \frac{100.00}{20} = 5.00 \text{ dollars}$$

2. **Calculate the Rolling Standard Deviation ($\sigma_{20,roll}$):**

- **Source of Values:** Sample standard deviation formula applied to the array of the last 20 residual spreads.
- Suppose the sample standard deviation calculation yields:

$$\sigma_{20,roll} = 4.00 \text{ dollars}$$

Step 4.4: Compute the Z-Score (Z_{20})

Apply the formal formula shown in the image:

$$Z_t = \frac{\hat{\epsilon}_t - \mu_{t,roll}}{\sigma_{t,roll}}$$

Substitute our numbers into the formula:

$$Z_{20} = \frac{20.00 - 5.00}{4.00}$$

$$Z_{20} = \frac{15.00}{4.00} = 3.75$$

Step 4.5: Trading Rule Evaluation

1. Compare $Z_{20} = 3.75$ against the formal trading rules:
 - Rule 1: Long spread when $Z_t < -2.0$.
 - Rule 2: Short spread when $Z_t > +2.0$.
2. Because $3.75 > +2.0$, the rule dictates **Short the Spread**.
3. **Execution Sizing:**
 - Sell short 1 share of Chevron (Y) at \$110.00.
 - Buy long 1.50 shares of ExxonMobil ($\hat{\beta} \times X$) at \$60.00 per share, total cost \$90.00.

Step 4.6: Trade Exit Calculation on Day 25

Five days later (Day 25), prices change:

- **Price of Chevron ($P_{Y,25}$):** \$102.00
- **Price of ExxonMobil ($P_{X,25}$):** \$63.00

1. **Calculate New Residual Spread ($\hat{\epsilon}_{25}$):**

$$\hat{\epsilon}_{25} = 102.00 - (1.50 \times 63.00) = 102.00 - 94.50 = 7.50 \text{ dollars}$$

2. **Retrieve Updated Rolling Parameters (Day 6 through Day 25):**

- Updated Rolling Mean ($\mu_{25,roll}$) = 5.50 dollars
- Updated Rolling Standard Deviation ($\sigma_{25,roll}$) = 4.00 dollars

3. **Calculate New Z-Score (Z_{25}):**

$$Z_{25} = \frac{7.50 - 5.50}{4.00} = \frac{2.00}{4.00} = 0.50$$

4. **Exit Check:**

- The Z-Score has moved from 3.75 down to 0.50, approaching zero ($Z_t \rightarrow 0$).
- The rule instructs to exit the trade.

5. **Profit Breakdown:**

- **Chevron Short Position Profit:** Sold short at \$110.00, bought back at \$102.00 $\rightarrow$ +\$8.00 profit.

- **ExxonMobil Long Position Profit:** Bought 1.50 shares at \$60.00 (\$90.00 total), sold at \$63.00 (\$94.50 total) → +\$4.50 profit.
- **Total Net Profit:** \$8.00 + \$4.50 = \$12.50.

PAIR SELECTION AND BACKTESTING

1. Plain English Explanation with Real-Life Analogy

Imagine you are standing in front of a locked door, and you want to find a key that opens it.

Scenario A: You are handed a single key by the building's master locksmith. You stick that one key into the lock, turn it, and the door opens on your very first try. That event is meaningful. Because you only tried one key, you can be fairly confident that the key was genuinely cut to fit that specific lock.

Scenario B: Now imagine you go to a giant junk yard and pick twenty random, rusty pieces of metal from a bucket. None of these keys were made for your door. You stand at the lock and try all twenty keys, one by one. Nineteen of them fail completely. But on key number fourteen, the metal happens to be slightly bent in just the right way that, with a little wiggling, the lock clicks open.

Did you find a master key that was designed for your door? No. You simply tried so many random pieces of trash metal that pure dumb luck produced one accidental fit.

If you test enough random items, chance alone will eventually make a completely useless item look like a perfect match. If you want to avoid being fooled by luck, you cannot just test random keys at random doors. You must limit your search to keys that actually belong to the same building, and you must calculate how many fake fits luck will produce on its own so you know whether your result is special or just random noise.

2. Formal Terms and Technical Jargon Linked to the Analogy

Here is how every part of that key story translates directly into formal finance and statistics terms:

- **Pair Selection / Trading Pairs:** In financial markets, instead of matching keys to locks, analysts look for two stock prices that move together in a stable, predictable relationship over time.
- **Hit-and-Trial Search:** This is the practice of grabbing random pairs of stocks out of thin air and testing them without a deliberate plan, exactly like picking twenty random rusty keys out of a junk bucket.
- **Significance Threshold (Five Percent Bar):** In statistics, a five percent threshold means you set a mathematical standard where a test result is only accepted if there is a five percent chance or less that random luck caused it. In our story, this is like using a lock where a random junk key only has a tiny five percent chance of turning the lock by accident.
- **The Multiple-Testing Problem:** This is the rule of probability stating that if you run many statistical tests, the overall chance of getting at least one lucky coincidence increases with every extra test you perform. Trying twenty keys inflates your total chance of seeing an accidental fit.
- **Spurious Regression / Spurious Cointegration:** This refers to a fake statistical relationship where two stock prices appear to move together tightly, but the connection is purely a temporary illusion created by random chance, just like the rusty piece of junk metal accidentally opening the door.
- **False Positive:** A false alarm where a test claims two things are connected when, in reality, they have no real connection.
- **Constrained Universe:** Instead of testing completely unrelated companies (like an airline company and a shoe manufacturer), you constrain your search universe by only testing companies within the exact same industry sector (such as two large banking corporations like Wells Fargo and Company). This raises the likelihood that any match you find is driven by genuine economic factors rather than random noise.

3. Practical Examples and Calculations

To determine how many fake matches you should expect strictly by luck, use the formula for expected false positive results:

$$\text{Expected False Positives} = N \times 0.05$$

Source of Every Value:

1. **N (Total Number of Tests):** Value is 20. **Source:** Explicitly stated in the image text under both the "Plain English" and "Formal" sections ("If you test twenty pairs...", "Test 20 random pairs").
2. **0.05 (Statistical Significance Threshold):** Value is 0.05 (which represents five percent). **Source:** Explicitly stated in the image text ("clears 5%", "at a 5% threshold").

Step-by-Step Calculation:

1. **Convert the percentage threshold into a decimal form:**

$$\text{Threshold} = \frac{5}{100} = 0.05$$

2. **Multiply the total number of tested pairs ($N = 20$) by the decimal threshold (0.05):**

$$\text{Expected False Positives} = 20 \times 0.05$$

3. **Perform the arithmetic:**

$$20 \times 0.05 = 1.0$$

Practical Meaning of the Result:

If you test 20 completely unrelated stock pairs using a five percent statistical threshold, you should mathematically expect roughly 1 false positive pair to pass your test purely by luck.

If your search through 20 random stock pairs yields exactly 1 passing pair, that passing pair is not a breakthrough discovery. It lands precisely on the baseline bar of pure random chance.

4. Conclusion

Testing stock pairs by hit-and-trial without adjusting for the number of tests performed guarantees that you will eventually be fooled by random noise. When you run multiple statistical tests, luck generates free false positives at a predictable rate calculated by multiplying your total test count by your threshold percentage. To reliably find true economic relationships rather than statistical illusions, you must first narrow your search to a logically constrained universe of stocks in the same sector, and then verify that your best result comfortably exceeds the baseline number of false matches that pure chance produces.

Further elaborate this passage "To reliably find true economic relationships rather than statistical illusions, you must first narrow your search to a logically constrained universe of stocks in the same sector, and then verify that your best result comfortably exceeds the baseline number of false matches that pure chance produces"

1. Plain English Explanation with Real-Life Analogy

Imagine you own a large farm and you want to find pairs of animals that are genuine best friends—meaning whenever one animal moves, the other animal follows it because they have a real bond (like a mother sheep and her lamb).

Here is the exact step-by-step process of how you find them without being fooled:

- **Step 1: Choose where you look.** If you pick one animal from the deep ocean (a fish) and one animal from the high sky (a eagle), they have zero relationship. If the eagle flies south at the exact same minute the fish swims south, that is pure coincidence. Searching the entire world at random wastes your time on pairs that cannot possibly be friends. Instead, you lock your search inside a single fenced sheep pen. Because all the sheep in that pen live together, eat together, and share the same mother, the chance of finding actual best friends inside that pen is extremely high.
- **Step 2: Set your test rule.** You decide to watch any chosen pair of animals for ten minutes. If they walk side-by-side for the whole ten minutes, you write down that they passed your test.
- **Step 3: Calculate what dumb luck will produce.** You know from past experience that if you test one hundred pairs of completely random animals that do not care about each other, pure luck will cause five of those random pairs to walk in the same direction for ten minutes anyway. Therefore, five passing pairs is your baseline for pure luck.
- **Step 4: Run your test and compare the counts.** You test one hundred pairs of sheep from inside the fenced pen.
 - If only five pairs pass your test, you have discovered nothing. Five is the exact number of fake fits that pure luck promised you.
 - If twenty-eight pairs pass your test, you have succeeded. Twenty-eight is far bigger than five. You know that five of those matches might be luck, but the remaining twenty-three pairs represent genuine, real friendships.

2. Formal Terms and Technical Jargon Linked to the Analogy

Here is how every step of that farm story connects to formal quantitative finance and statistics terms:

- **Logically Constrained Universe:** This is the fenced sheep pen. Instead of scanning all thousands of stocks in the entire stock market, you constrain your search to a specific group of companies that operate in the exact same economic sector (for example, comparing commercial banks like Wells Fargo and Company with JPMorgan Chase and Company).
- **True Economic Relationship (Cointegration):** This is the mother sheep and her lamb. It describes two stock prices that are bound together by shared market fundamentals (such as interest rates, raw material costs, or customer bases). Even if their prices temporarily drift apart, the underlying economic link forces them back together over time.
- **Statistical Illusion (Spurious Correlation):** This is the fish and the eagle moving south at the same time. It occurs when two totally unrelated stock prices happen to move in the same direction for a short period due to random coincidence, even though they share no common business drivers.
- **Baseline Number of False Matches (Expected False Positives):** This is the calculation showing that five random pairs will pass the test purely by luck. It represents the mathematical number of false signals you expect to see when running multiple statistical tests at a specific significance threshold.
- **Beating the Baseline (Exceeding Expected Noise):** This is getting twenty-eight passing pairs instead of just five. In quantitative research, you do not just celebrate finding a single passing stock pair; you check whether the total number of passing pairs across your entire search is significantly higher than what random chance predicts.

3. Practical Examples and Calculations

To demonstrate how narrowing the universe and beating the baseline works in practice, we look at two mathematical search scenarios using the expected false positive formula:

$$\text{Expected False Positives} = \text{Total Pairs Tested} \times \text{Statistical Significance Threshold}$$

Source of Every Single Value:

1. **Total Pairs Tested (N) = 100:** A standardized sample size chosen to represent testing one hundred stock combinations.
2. **Statistical Significance Threshold (α) = 0.05 (Five Percent):** Derived directly from the standard five percent statistical cutoff referenced in the source text.
3. **Expected False Matches = 5:** Calculated directly by multiplying 100 by 0.05.
4. **Observed Passing Pairs (K):**
 - Scenario A Observed Value (5): A test outcome showing a failed, unconstrained search.
 - Scenario B Observed Value (28): A test outcome showing a successful, constrained search.

Scenario A: The Flawed Search (Unconstrained Universe)

You test 100 pairs made of completely random, unrelated stocks across different industries (for example, pairing a gold mining company with a fast-food restaurant chain).

1. **Calculate expected false matches from pure chance:**

$$\text{Expected False Matches} = 100 \times 0.05 = 5$$

2. **Count observed passing pairs:**

$$\text{Observed Passing Pairs} = 5$$

3. **Calculate net genuine relationships:**

$$\text{Net Genuine Pairs} = \text{Observed Passing Pairs} - \text{Expected False Matches}$$

$$\text{Net Genuine Pairs} = 5 - 5 = 0$$

- **Result:** Every single passing pair in this search is a statistical illusion caused by random market noise.

Scenario B: The Proper Search (Constrained Universe)

You narrow your search universe to 100 pairs composed strictly of companies within the exact same economic sector (for example, pairing regional commercial banks with other regional commercial banks).

1. **Calculate expected false matches from pure chance:**

$$\text{Expected False Matches} = 100 \times 0.05 = 5$$

2. Count observed passing pairs:

$$\text{Observed Passing Pairs} = 28$$

3. Calculate net genuine relationships:

$$\text{Net Genuine Pairs} = \text{Observed Passing Pairs} - \text{Expected False Matches}$$

$$\text{Net Genuine Pairs} = 28 - 5 = 23$$

- **Result:** The observed number of passing pairs (28) comfortably exceeds the random baseline (5). You can be confident that approximately 23 of these pairs reflect real economic relationships rather than statistical noise.

4. Conclusion

Searching for trading pairs without constraining your search universe is like fishing for economic relationships in random noise: luck will always hand you a small handful of fake matches that clear your statistical bar. By restricting your search to companies in the same industry sector, you raise the underlying odds that any price connection is driven by real economic forces. Furthermore, by calculating your baseline rate of false positives beforehand, you ensure that you only trade when your total number of passing pairs vastly outnumbers the accidental fits generated by pure chance.

Walk-forward estimation

1. Real-Life Simple English Analogy

Imagine you are trying to guess how many carrots your donkey will eat today based on how many miles it walked. You want to test how good your guessing rule is over an entire past year.

The Cheating Way (Static Estimation):

You wait until the entire year is over. You take all 365 days of data, look at the whole year at once, and calculate one single average rule: *"For every 1 mile walked, the donkey eats 2 carrots."*

Then, you go back to evaluate how you would have performed on Day 10. You use that single rule (*2 carrots per mile*) to make your guess on Day 10.

Why is this cheating? Because on Day 10 in real life, you could **not** have known what happened in July or December! In July, a heatwave occurred, and in December, a blizzard occurred. By using a single rule calculated from the whole year, your Day 10 guess secretly uses future information that you did not possess on Day 10.

The Honest Way (Walk-Forward Estimation):

If you want to know how smart you *actually* would have been on Day 10, you are only allowed to look backward at the days before Day 10 (for example, Days 7, 8, and 9). You calculate your carrot-guessing rule using only those past 3 days.

When you move to Day 11, you drop Day 7 and use Days 8, 9, and 10 to update your rule for Day 11. As you walk forward day by day, you look through a moving window of past history. You never look ahead into tomorrow.

2. Technical Terms and Link to the Analogy

Here is how the mathematical terms from the image connect directly to the donkey analogy:

- **Time Step (t):** The current day or exact point in time where you stand right now (for example, Day 10).
- **Estimated Beta Hat ($\hat{\beta}$):** The multiplier rule that links two variables together. In the analogy, $\hat{\beta}$ is the rule: *"How many carrots per mile walked."* In trading, it measures how much Asset A moves when Asset B moves by one dollar.
- **Global Sample $[t_0, T]$:** The entire dataset from start time t_0 to end time T (the whole 365-day year). Fitting a static $\hat{\beta}$ on this range means calculating one single multiplier using future data you shouldn't have access to yet.
- **Rolling Window Size (w):** The fixed number of past days you look back at (for example, a lookback window of $w = 3$ days).
- **Walk-Forward Estimation ($\hat{\beta}_t$ fit on $[t - w, t)$):** Calculating a fresh beta hat specifically at time t using only data from $t - w$ up to $t - 1$, strictly excluding any data at or after time t ($\geq t$).

- **Spread (S_t):** The gap between the actual price of Asset A and expected price predicted by Asset B using beta hat: $S_t = \text{Price}_A(t) - (\hat{\beta}_t \times \text{Price}_B(t))$. This represents how many extra or fewer carrots the donkey ate compared to the rule's prediction.
- **Z-Score (Z_t):** A measurement of how unusual today's spread is compared to the average spread over the recent past window, measured in standard deviations.

3. Practical Application to Stocks with Detailed Step-by-Step Calculations

Let us apply Walk-Forward Estimation to pairs trading between two stocks: **Stock A** and **Stock B**.

The Dataset

Suppose we have daily closing prices for 4 consecutive days:

- **Day 1 ($t = 1$):** Stock A = \$10, Stock B = \$5
- **Day 2 ($t = 2$):** Stock A = \$14, Stock B = \$6
- **Day 3 ($t = 3$):** Stock A = \$15, Stock B = \$7
- **Day 4 ($t = 4$):** Stock A = \$12, Stock B = \$9

We set our rolling lookback window size to $w = 3$ **days**. We stand at **Day 4 ($t = 4$)** and want to compute our walk-forward estimate $\hat{\beta}_4$, the spread S_4 , and the Z-score Z_4 .

Step 1: Isolate the Lookback Window Data

For Day $t = 4$ with window $w = 3$, our window covers $[4 - 3, 4) = [1, 4)$, which means **Days 1, 2, and 3**. Data from Day 4 itself (and any future days) is strictly excluded from calculating $\hat{\beta}_4$.

Data inside our window:

- Day 1: $P_{A,1} = 10, P_{B,1} = 5$
- Day 2: $P_{A,2} = 14, P_{B,2} = 6$
- Day 3: $P_{A,3} = 15, P_{B,3} = 7$

Step 2: Calculate the Walk-Forward Estimate $\hat{\beta}_4$

The statistical formula for linear regression slope ($\hat{\beta}$) is:

$$\hat{\beta}_t = \frac{\sum (P_B - \bar{P}_B)(P_A - \bar{P}_A)}{\sum (P_B - \bar{P}_B)^2}$$

1. **Calculate the average price of Stock B ($\bar{P}_B$) in the window:**

$$\bar{P}_B = \frac{5 + 6 + 7}{3} = \frac{18}{3} = 6.0$$

2. **Calculate the average price of Stock A ($\bar{P}_A$) in the window:**

$$\bar{P}_A = \frac{10 + 14 + 15}{3} = \frac{39}{3} = 13.0$$

3. **Calculate price deviations from the average for each past day:**

- **Stock B deviations ($P_B - \bar{P}_B$):**
 - Day 1: $5 - 6.0 = -1.0$
 - Day 2: $6 - 6.0 = 0.0$
 - Day 3: $7 - 6.0 = +1.0$
- **Stock A deviations ($P_A - \bar{P}_A$):**
 - Day 1: $10 - 13.0 = -3.0$

- Day 2: $14 - 13.0 = +1.0$
- Day 3: $15 - 13.0 = +2.0$

4. **Calculate Covariance Sum (numerator):**

$$\text{Covariance Sum} = (-1.0 \times -3.0) + (0.0 \times 1.0) + (1.0 \times 2.0) = 3.0 + 0.0 + 2.0 = 5.0$$

5. **Calculate Variance Sum of Stock B (denominator):**

$$\text{Variance Sum} = (-1.0)^2 + (0.0)^2 + (1.0)^2 = 1.0 + 0.0 + 1.0 = 2.0$$

6. **Divide Covariance Sum by Variance Sum to get $\hat{\beta}_4$:**

$$\hat{\beta}_4 = \frac{5.0}{2.0} = 2.5$$

Step 3: Calculate Historical Spreads in the Lookback Window

Now, evaluate what the spread was on each day inside the past window using our newly calculated $\hat{\beta}_4 = 2.5$:

$$\text{Spread} = P_A - (\hat{\beta}_4 \times P_B)$$

- **Day 1 Spread (S_1):** $10 - (2.5 \times 5) = 10 - 12.5 = -2.5$
- **Day 2 Spread (S_2):** $14 - (2.5 \times 6) = 14 - 15.0 = -1.0$
- **Day 3 Spread (S_3):** $15 - (2.5 \times 7) = 15 - 17.5 = -2.5$

Step 4: Calculate Mean (μ_S) and Standard Deviation (σ_S) of Past Spreads

1. **Mean of historical spreads (μ_S):**

$$\mu_S = \frac{-2.5 + (-1.0) + (-2.5)}{3} = \frac{-6.0}{3} = -2.0$$

2. **Deviations of past spreads from the mean ($\mu_S = -2.0$):**

- Day 1: $-2.5 - (-2.0) = -0.5$
- Day 2: $-1.0 - (-2.0) = +1.0$
- Day 3: $-2.5 - (-2.0) = -0.5$

3. **Squared deviations:**

- Day 1: $(-0.5)^2 = 0.25$
- Day 2: $(1.0)^2 = 1.00$
- Day 3: $(-0.5)^2 = 0.25$

4. **Sum of squared deviations:**

$$0.25 + 1.00 + 0.25 = 1.50$$

5. **Sample Variance (s^2) with $n - 1 = 3 - 1 = 2$ degrees of freedom:**

$$s^2 = \frac{1.50}{2} = 0.75$$

6. **Sample Standard Deviation (σ_S):**

$$\sigma_S = \sqrt{0.75} \approx 0.866$$

Step 5: Calculate Today's Spread (S_4) and Z-Score (Z_4) at Day 4

Today is Day 4. Stock A price is \$12 and Stock B price is \$9.

1. **Calculate Today's Spread (S_4):**

$$S_4 = P_{A,4} - (\hat{\beta}_4 \times P_{B,4}) = 12 - (2.5 \times 9) = 12 - 22.5 = -10.5$$

2. Calculate Today's Z-Score (Z_4):

$$Z_4 = \frac{S_4 - \mu_S}{\sigma_S} = \frac{-10.5 - (-2.0)}{0.866} = \frac{-8.5}{0.866} \approx -9.815$$

Why This Matters (The Cheating vs. Honest Comparison)

- **Walk-Forward Result:** Our Z-score is -9.815 , indicating an extreme deviation based exclusively on past information.
- **If we had cheated (Static Global Method):** We would have included prices from Day 4 (and subsequent future days) into calculating a single global beta. That future information would artificially flatten the historical error, making the backtest appear unrealistically profitable because the model "knew" where prices were heading next.

RISK METRICS

1. Real-Life Simple English Analogy

Imagine you own a donkey and a wooden cart that carries bags of carrots along a bumpy, uneven dirt trail. You want to evaluate whether your cart-driving strategy is profitable, safe, and reliable over time. To figure this out, you ask four simple questions about your daily journey:

1. How much money did you actually make today based on yesterday's plan?

Suppose yesterday afternoon you set a rule: *"If the trail tilted left yesterday, today I will steer right."* You cannot use today's trail tilt to make today's decision because you do not know today's tilt until you are already standing on it. Therefore, today's actual gain depends entirely on yesterday's decision combined with today's real movement of the trail.

2. How smooth was your profit compared to how bumpy the ride was?

If you made 100 coins in profit, but the cart violently shook back and forth so hard that half your carrots almost spilled out every five minutes, that is a risky strategy. To test smoothness, you divide your average daily coin earnings by how wildly your daily earnings bounced around, and then scale that number up to a full year of trail rides to get a single performance score.

3. What was your worst nightmare fall during the journey?

Imagine tracking your total stash of gold coins over time. At your absolute highest point, you had 100 gold coins. Then you hit a bad stretch of road and your stash dropped down to 70 gold coins before it began recovering. That drop from your peak of 100 coins down to the valley of 70 coins is a loss of 30 gold coins (a 30 percent decline). That deep hole represents your worst historical drop.

4. How long does it take for a pushed cart to drift back to the middle of the road?

Suppose a sudden gust of wind pushes your cart off to the edge of the dirt path. Naturally, the cart tends to roll back toward the center groove of the road over time. If the cart gets pushed 10 inches off to the side, how many days does it take for that 10-inch push to shrink down to 5 inches (halfway back)? Knowing how fast it returns halfway tells you whether your steering speed matches the natural rhythm of the road.

2. Technical Terms and Formal Concepts Linked to the Analogy

Here is how those cart-and-carrot ideas translate into formal quantitative finance terms from the text:

A. Daily Spread Profit and Loss (Profit and Loss_t)

- **Formula:**

$$\text{Profit and Loss}_t = \text{signal}_{t-1} \times (\Delta Y_t - \hat{\beta}_{t-1} \Delta X_t)$$

- **Link to Analogy:** This calculates how many coins you earned on day t .

- **Component Definitions:**

- signal_{t-1} : Your position size and direction decided at the end of yesterday (day $t - 1$). You cannot trade on today's signal (signal_t) because today's market data is not available until after the market closes.
- ΔY_t : The daily price change of Asset Y from yesterday to today ($\Delta Y_t = Y_t - Y_{t-1}$).
- ΔX_t : The daily price change of Asset X from yesterday to today ($\Delta X_t = X_t - X_{t-1}$).
- $\hat{\beta}_{t-1}$ (Beta hat): The estimated hedge ratio calculated using historical data up to yesterday. It dictates how many units of Asset X you sell short for every unit of Asset Y you buy, ensuring market-wide movements are canceled out.

- $(\Delta Y_t - \hat{\beta}_{t-1} \Delta X_t)$: The daily net change in the hedged price difference (the spread) between Asset Y and Asset X.

B. Annualized Sharpe Ratio

- **Formula:**

$$\text{Sharpe Ratio} = \frac{\bar{r}}{\sigma_r} \sqrt{252}$$

- **Link to Analogy:** This is your annual smoothness score.
- **Component Definitions:**
 - $\bar{r}$ (r -bar): The mean (average) daily return generated by the strategy.
 - σ_r (σ sub r): The standard deviation (volatility) of daily returns.
 - $\sqrt{252}$: The square root of 252, representing the standard number of open stock trading days in one calendar year. Multiplying by $\sqrt{252}$ converts daily risk-adjusted returns into an annualized scale.

C. Maximum Drawdown

- **Definition:** The largest percentage or dollar decline from a historical peak to the subsequent lowest trough before a new peak is achieved on your cumulative Profit and Loss curve.
- **Link to Analogy:** This measures the drop in your stash of coins from its peak down to its lowest valley.

D. Half-Life of Mean Reversion

- **Formula:**

$$\text{half-life} = \frac{-\ln(2)}{\ln(\phi)}$$

- **Link to Analogy:** This measures the number of days required for an abnormal price stretch to decay halfway back to its historical average.
- **Component Definitions:**
 - $\epsilon_t = \phi \epsilon_{t-1} + u_t$: An Autoregressive process of order 1, denoted as AR(1), where ϵ_t represents the spread deviation on day t .
 - ϕ (Phi): The autoregressive parameter (persistence coefficient), measuring how much of yesterday's deviation (ϵ_{t-1}) carries over into today.
 - u_t : The random noise or unexpected shock on day t .
 - $\ln$: The natural logarithm function.

3. Practical Application with Stocks and Detailed Step-by-Step Calculations

Consider a pairs trading strategy between two stock market assets: **Coca-Cola Company (Stock Y)** and **PepsiCo Incorporated (Stock X)**.

Calculation 1: Daily Spread Profit and Loss (Profit and Loss_t)

Given Input Data:

- Yesterday's Signal (signal_{t-1}) = +1.50 (long 1.50 units of the spread).
- Yesterday's Estimated Hedge Ratio ($\hat{\beta}_{t-1}$) = 0.80.
- Yesterday Coca-Cola price = \$60.00, Today Coca-Cola price = \$62.00.
 - Change in Stock Y (ΔY_t) = \$62.00 – \$60.00 = +\$2.00.
- Yesterday PepsiCo price = \$170.00, Today PepsiCo price = \$171.00.
 - Change in Stock X (ΔX_t) = \$171.00 – \$170.00 = +\$1.00.

Formula:

$$\text{Profit and Loss}_t = \text{signal}_{t-1} \times (\Delta Y_t - \hat{\beta}_{t-1} \Delta X_t)$$

Step-by-Step Arithmetic:

1. Multiply yesterday's hedge ratio by the change in PepsiCo's price:

$$\hat{\beta}_{t-1} \times \Delta X_t = 0.80 \times 1.00 = 0.80$$

2. Subtract this hedged change from Coca-Cola's price change to find the net spread move:

$$\Delta Y_t - \hat{\beta}_{t-1} \Delta X_t = 2.00 - 0.80 = 1.20$$

3. Multiply the net spread move by yesterday's signal:

$$\text{Profit and Loss}_t = 1.50 \times 1.20 = 1.80$$

Result: Today's Profit and Loss is **\$1.80** per trading unit.

Calculation 2: Annualized Sharpe Ratio

Given Input Data:

- Mean daily return ($\bar{r}$) = 0.0010 (0.10% return per day).
- Standard deviation of daily returns (σ_r) = 0.0080 (0.80% volatility per day).
- Trading days in a year = 252.

Formula:

$$\text{Sharpe Ratio} = \frac{\bar{r}}{\sigma_r} \sqrt{252}$$

Step-by-Step Arithmetic:

1. Divide the average daily return by the daily standard deviation:

$$\frac{\bar{r}}{\sigma_r} = \frac{0.0010}{0.0080} = 0.125$$

2. Calculate the square root of 252:

$$\sqrt{252} \approx 15.87450786$$

3. Multiply the daily ratio by the square root of 252:

$$\text{Sharpe Ratio} = 0.125 \times 15.87450786 = 1.98431348$$

Result: The Annualized Sharpe Ratio is **1.98** (indicating high risk-adjusted efficiency).

Calculation 3: Maximum Drawdown

Given Input Data (Cumulative Profit and Loss over 5 consecutive days):

- Day 1 Cumulative Profit and Loss = \$100.00
- Day 2 Cumulative Profit and Loss = \$120.00 (Historical Peak)
- Day 3 Cumulative Profit and Loss = \$105.00
- Day 4 Cumulative Profit and Loss = \$90.00 (Lowest Trough)
- Day 5 Cumulative Profit and Loss = \$115.00 (Recovery)

Step-by-Step Arithmetic:

1. Identify the highest historical peak achieved before the decline:

$$\text{Peak} = \$120.00$$

2. Identify the lowest trough reached following that peak before making a new high:

$$\text{Trough} = \$90.00$$

- Calculate the total dollar decline from peak to trough:

$$\text{Dollar Loss} = \text{Peak} - \text{Trough} = \$120.00 - \$90.00 = \$30.00$$

- Calculate the percentage decline relative to the peak:

$$\text{Maximum Drawdown Percentage} = \frac{\text{Peak} - \text{Trough}}{\text{Peak}} \times 100$$

$$\text{Maximum Drawdown Percentage} = \frac{30.00}{120.00} \times 100 = 0.25 \times 100 = 25.00\%$$

Result: The Maximum Drawdown is **\$30.00** or **25.00%**.

Calculation 4: Half-Life of Mean Reversion

Given Input Data:

- Fitting an Autoregressive process of order 1 model ($\epsilon_t = \phi \epsilon_{t-1} + u_t$) yields an estimated coefficient $\phi = 0.85$.
- Standard natural logarithm constants: $\ln(2) \approx 0.69314718$.

Formula:

$$\text{half-life} = \frac{-\ln(2)}{\ln(\phi)}$$

Step-by-Step Arithmetic:

- Compute the natural logarithm of the autoregressive parameter ϕ (0.85):

$$\ln(0.85) \approx -0.16251893$$

- Take the negative natural logarithm of 2:

$$-\ln(2) = -0.69314718$$

- Divide the negative natural log of 2 by the natural log of ϕ :

$$\text{half-life} = \frac{-0.69314718}{-0.16251893} \approx 4.2649$$

Result: The half-life of mean reversion is **4.26 days**.

Practical Interpretation: An abnormal divergence in the price spread between Coca-Cola and PepsiCo takes approximately 4.26 trading days to decay halfway back to its baseline equilibrium. Because 4.26 days is much shorter than a 20-day statistical Z-score window, the 20-day lookback window is appropriate because it captures multiple full decay cycles without lagging behind the true rhythm of the spread.